Qijun (Sergiu) Han

Computer Science and Engineering Ph.D. Student · UC Santa Cruz
ghan19@ucsc.edu · Santa Cruz, CA · GitHub · Research blog

Research interests include computer-use agents, GUI/CLI orchestration, and long-horizon evaluation.

EDUCATION

University of California, Santa Cruz Fall 2026 - Present
Ph.D. in Computer Science and Engineering. Advised by Prof. Cihang Xie.

Southern University of Science and Technology Aug 2022 - Jun 2026
B.Eng. in Automation

SELECTED PREPRINT

VLAA-GUI: Knowing When to Stop, Recover, and Search, A Modular Framework for GUI Automation
Qijun Han*, Haoqin Tu*, Zijun Wang, et al. (*equal contribution) · 2026
arXiv:2604.21375 · Project and demos · Code

RESEARCH EXPERIENCE

Computer-Use Agent Research · VLAA Lab, UC Santa Cruz Jul 2025 - Present
- Co-developed a model-agnostic desktop-agent harness with completion verification, loop recovery, workflow search, and GUI and programmatic actions.
- Ran OSWorld component ablations with Ray-managed VM rollouts and analyzed execution traces for premature success claims and repeated actions.
- VLAA-GUI achieved 77.5% task success on OSWorld-Verified and 61.0% on WindowsAgentArena with a 100-step budget, using the best backbone for each benchmark.

ADRec · Undergraduate Thesis, SUSTech 2026
- Built a generative recommender with Sentence-T5 embeddings, a Transformer history encoder, and rectified-flow sampling. Measured Recall@K and NDCG@K on Amazon Reviews 2014 subsets.

Corrective Permutations · Vision Research, SUSTech Oct 2024 - Sep 2025
- Used linear probes, t-SNE, and Grad-CAM to investigate why spatial rearrangements of image patches can correct ViT and CNN misclassifications.

SELECTED ENGINEERING EXPERIENCE

RoboMaster Autonomous Navigation · ARTINX, SUSTech Mar 2024 - Aug 2024
- Led autonomous decision logic with BehaviorTree.CPP and implemented ROS2/Nav2 navigation with Fast-LIO2 odometry. The team placed eighth regionally and in the national top 32.

Wheeled-Legged Robot Navigation · SUSTech Apr 2025 - Apr 2026
- Co-inventor on Chinese patent application 202610614507.9 for RL-based wheeled-legged robot navigation and transport, filed May 2026.

TECHNICAL SKILLS

Programming Python, C/C++
Machine learning PyTorch, Hugging Face Transformers
Agent development Agent orchestration, LLM tool calling, GUI/CLI automation
Evaluation OSWorld benchmarking, ablation studies, trajectory analysis
Infrastructure Linux, Ray, Docker, QEMU/KVM, Git, Slurm